LLMs Unplugged | Cybernetic Studio

Australian National University

Synthetic Data

Use your language model to generate new training data, then train a new model on that synthetic data to see how patterns degrade or change.

You will need

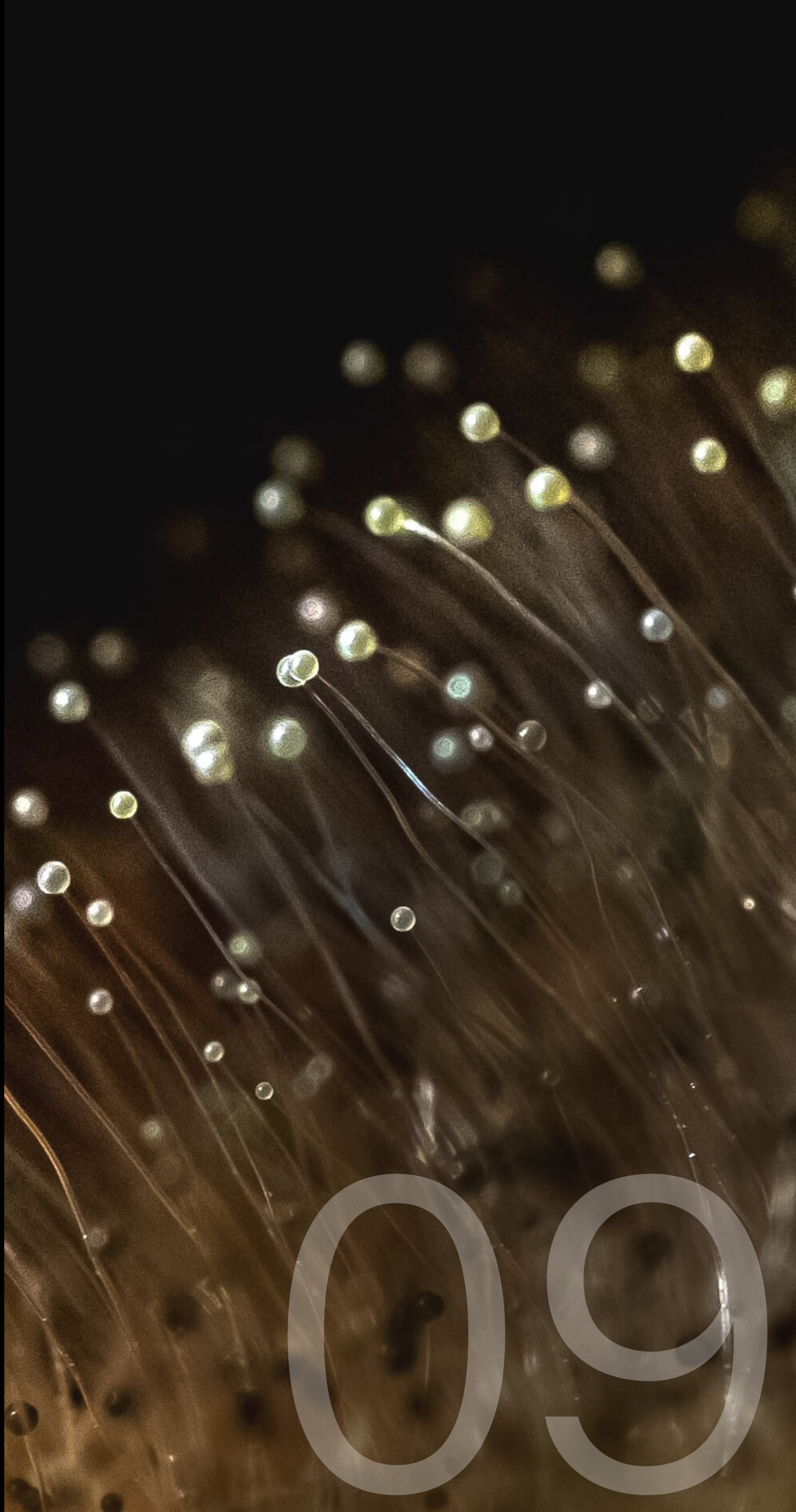
- a completed model from an earlier lesson
- pen, paper & dice for text generation
- grid paper for a new model

Your goal

To generate synthetic text using your model, then train a new “generation 2” model on that synthetic output. Compare the two models to observe what patterns are preserved or lost. **Stretch goal**: train a generation 3 model on generation 2 output. Or go “full Joker”.

Key idea

Models trained on synthetic data (output from other models) can drift from the original patterns. This demonstrates model collapse and the importance of real training data.



Algorithm

1. **generate synthetic text**:
   - use your existing model to generate text (as in Basic Generation)
   - generate enough text for meaningful training (at least 50-100 words)
   - this is your *synthetic training corpus*
2. **train generation 2 model**:
   - create a new grid following the Basic Training algorithm
   - use your synthetic text as the input corpus
   - this new model learns from AI-generated text, not human-written text
3. **compare the models**:
   - look for words that appear in the original but not in generation 2
   - compare the counts for cells that appear in both
   - generate text from both models and compare the outputs

Example

Original training text: *"See Spot run. See Spot jump."*

Generation 1 model's synthetic output: *"See run. Run spot. Spot run run."*

Notice how the synthetic text:
- uses all the same words as the original
- has different patterns (more `run` `run`, no `spot` `jump`)
- might lose some variety from the original

Generation 2 model trained on the synthetic output will amplify these changes:
- `run` `run` becomes more common
- `spot` `jump` disappears entirely
- new unlikely patterns may emerge

Joker mode

Instead of generating synthetic text from an existing model, create a completely random model:

1. draw a grid with any words you choose in the rows and columns
2. add tally marks in any cells you want, with any frequencies
3. this creates a model with no connection to real text patterns
4. generate text from this random grid using dice as normal
5. train a generation 2 model on the output from your random grid

Example

A completely random Joker grid might look like this:

	`pizza`	`robot`	`moon`	`dance`
`pizza`	3		1	2
`robot`	1	4		1
`moon`		2	1	3
`dance`	2	1	2	